

## The 66th ASH Annual Meeting Abstracts

## POSTER ABSTRACTS

## 701.EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

**Long-Lasting AML-Specific Antibody Responses after Allogeneic Hematopoietic Progenitor Cell Transplantation**

Rogers Nahui Palomino, PhD<sup>1</sup>, Etsuko Yasuda, PhD<sup>2</sup>, Nienke Haverkate, MSC<sup>2</sup>, Hergen Spits, PhD<sup>2</sup>, Tim Beaumont, PhD<sup>3</sup>, Bianca Blom, PhD<sup>2</sup>, Mette D. Hazenberg, MD PhD<sup>2,4,5</sup>

<sup>1</sup>Department of Experimental Immunology, Amsterdam University Medical Center, University of Amsterdam, Amsterdam, Netherlands

<sup>2</sup>Department of Experimental Immunology, Amsterdam University Medical Center, University of Amsterdam, Amsterdam, Netherlands

<sup>3</sup>Department of Medical Microbiology and Infection Prevention, Amsterdam University Medical Center, University of Amsterdam, Amsterdam, Netherlands

<sup>4</sup>Department of Hematopoiesis, Sanquin Research, Amsterdam, Netherlands

<sup>5</sup>Department of Hematology, Amsterdam University Medical Center, University of Amsterdam, Amsterdam, Netherlands

**Background:** Allogeneic hematopoietic progenitor cell transplantation (HCT) can cure patients with hematologic malignancies such as acute myeloid leukemia (AML) by inducing durable graft versus leukemia (GvL) immune responses. However, this procedure is often complicated by graft versus host disease and opportunistic infections. The development of less toxic immunotherapies such as chimeric antigen receptor (CAR) T or bispecific T cell engager therapies is hindered by the lack of unique tumor-specific antigens. In patients with AML, who were cured after allogeneic HCT, we demonstrated the presence of antibody producing B cells that target the U5 snRNP200 complex (Gillissen et. al. Blood 2018). In non-malignant cells, the U5 snRNP200 complex is a nuclear protein that is part of the spliceosome. We found that in 30-50% of AML patients, U5 snRNP200 is expressed on the surface of AML blasts but not on normal hematopoietic stem cells. This finding was recently confirmed in an independent cohort (Knorr et. al. Nat Cancer 2023). Here, we investigated the durability of the U5 snRNP200 antibody response in the same patients, who are now over 10 years after allogeneic HCT.

**Methods:** Characteristics of the 2 patients, who remain in complete remission 12 years after allogeneic HCT for myelomonocytic AML, were described previously (Gillissen et.al. Blood 2018, pts 58 and 59). Memory IgG B-cells (CD19+, CD27+, IgM-, IgA-) were sorted from peripheral blood cells, immortalized by overexpression of BCL6 and Bcl-xL, seeded 10-20 cells/well, and cultured for 2-3 weeks. B-cell supernatants containing secreted antibodies were screened for the presence of U5 snRNP200 antibodies by snRNP200 protein based fluorescent ELISA and assessed for their capacity to bind AML cell lines THP-1, Molm13, SH2, and U937. Monoclonal antibodies were retrieved after single cell culture, sequenced and heavy and light chain domains cloned. To study cross-reactivity to commensal bacteria, U5 snRNP200 antibodies were incubated with bacteria derived from frozen stool of healthy donors and the patients.

**Results:** Blood samples were collected from patients cured of AML in whom U5 snRNP200 antibody producing B cells were identified 10 years ago, 2 years after allogeneic HCT. These patients have remained AML-free, for over 12 years after transplantation. Screening of their current B cell repertoire identified 6 B cell clones producing U5 snRNP200 antibodies, as tested by U5 snRNP200 ELISA. The specificity of these antibodies to AML cells was confirmed by binding to the AML cell lines THP-1, Molm13, SH2, and U937 via FACS. Sequencing of these antibodies demonstrated involvement of different IGHV families and a higher number of somatic hypermutations in their CDR3s compared to the U5 snRNP200 antibodies identified 10 years ago. U5 snRNP200 antibodies have not been detected in healthy individuals or in patients receiving allogeneic HCT for hematologic malignancies other than AML. The sustained presence of U5 snRNP200 specific B-cell immunity in these patients for years, presumably long after AML clearance, suggests cross-reactivity with another factor. This, combined with findings from checkpoint inhibitor and CAR T cell studies indicating that tumor immunity is co-dependent on microbiome related factors prompted us to study cross-reactivity of U5 snRNP200 antibodies to commensal bacteria. Five out of 6 U5 snRNP200 antibodies bind commensal bacteria, while isotype controls and RSV or SARS-CoV-2 specific antibodies did not, as measured by FACS. We are currently identifying the specific bacterial strains to which these antibodies are cross-reacting.

**Conclusions:** U5 snRNP200 is a tumor specific antigen expressed on the surface of AML cells. Allogeneic HCT recipients who successfully cleared AML generated U5 snRNP200-specific (donor) B-cells early after transplantation, and these B cell re-

sponses persisted for more than a decade. Cross-reactivity with commensal bacteria may contribute to this long-lasting immune response. Identification of AML-specific antibodies from allogeneic HCT recipients who successfully cleared AML helps discover novel targets for AML treatment. Additionally, U5 snRNP200 antibodies can be developed into therapeutic formats, such as CAR-T cells and bispecific antibodies.

**Disclosures**

HS, TB and MDH hold a patent covering human U5 snRNP antibodies (PCT/NL2014/050873).

**Disclosures** No relevant conflicts of interest to declare.

<https://doi.org/10.1182/blood-2024-207046>